WM-D6C Servo Replacement Module — System Block Diagram

Dual Variant Architecture | STM32G0B1KBU6 | Open Source Hardware

VARIANT A — USB-C Power Delivery

VARIANT B — Barrel Jack (Polarity-Agnostic)

Shared core shown in BLUE • Variant-specific sections colour-coded • WM-D6C machine interface shown in DARK BLUE

VARIANT A — USB-C PD INPUT

SHARED CORE — STM32G0B1KBU6 + POWER MANAGEMENT

VARIANT B — BARREL JACK INPUT

USB-C Receptacle (Panel / Flex tail) VBUS D+ D− CC1 CC2 GND ▶ **IP2721 PD Trigger 9V PDO Request** ▶ **TPS62xx Buck Reg 9V → 6.0V 600mA** ▶ **MT3608 Boost Conv 6V → 10.8V 500mA** ▶ **MCP1700 LDO 6V → 3.3V 250mA**

Barrel Jack CN301 Panel (Retained) ±Polarity In

CC1 / CC2 lines to STM32 UCPD for PD contract monitoring VBUS → IP2721 → 9V negotiated D+/D− → ESD → STM32 USB PHY

▼ 6V regulated B+1 rail to WM-D6C machine • ▼ 10.8V B+3 motor rail • ▼ 3.3V to STM32 VDD

Polyfuse 500mA Ovrcurrent ▶ ◀ **Ideal Diode Bridge (LTC4359) Polarity OK**

USB ESD Protection USBLC6-2SC6Y D+ / D− clamp CC Lines to PA8 / PB15 (UCPD1) VBUS Monitor ADC input ▶

STM32G0B1KBU6

UFQFPN32 • 64 MHz Cortex-M0+ • 128KB Flash • 144KB RAM

TIM2_CH1 PA0 FG Pulse Capture Input	**ADC_IN PA1/2/3 RV601 RV602 RV603**	**DAC1_OUT PA4 Motor Drive Analog Output**	**GPIO PA5/7 Motor EN S601 Speed Sw**
	Digital PLL PI Servo Loop Fixed-point C ISR < 500ns	**USB CDC Live Tuning Firmware DFU no ST-Link**	**UCPD1 PA8/PB15 CC1/CC2 PD Ctrl**
TIM3_CH1 PA6 PWM Out (Opt B)	**USART1 PA9 Debug TX Test Pad**	**PA11/12 USB D−/D+ FS 12Mbit Crystal-less**	**SWD PA13/14 SWDIO SWDCLK**

◀ TVS Clamp SMBJ7.0A 6.5V standoff Transient prot 220µF Bulk Input Filter 6V Clean Rail → Shared Core

▼ WM-D6C MACHINE INTERFACE ▼

FG901 Optical Sensor FG_RAW → PA0 Level shift + BAT54 clamp	Q601 (2SB733) PNP Motor Drv DAC_OUT → Base NPN level shift (Option A or B)	Speed Pots RV601 → PA1 RV602 → PA2 RV603 → PA3 ADC_IN1/2/3	Motor Rail CP304 / MT3608 10.8V B+3 → Q703/Q704 → M901 Motor	S601 Speed Sw → PA7 GPIO Motor Enable IC601 pin 7 → PA5	B+1 Rail (6V) → Machine audio → Logic circuits → Auto-Off board GND (Chassis)

STM32G0B1KBU6 • UFQFPN32 • Bare-metal C firmware • Replaces CP304 + CX20084 + CN301